ChE 312 – Transport Processes II

I. Equilibrium separations: Distillation.

I.d. Getting real: Practical aspects of distillation..

- What can we do with this understanding? Lots!
- Explore effects of specification changes.
- $R \rightarrow \infty$ (total reflux) and R_{min} and their usefulness as bounds.
- Optimal R vs R_{min}: Economic analysis of utility costs.
- Overall and tray efficiencies.
- Determining column height; determining diameter by flooding analysis.
- Remark on enthalpy-concentration analysis for L/V changes.
- Consider column configuration variations.

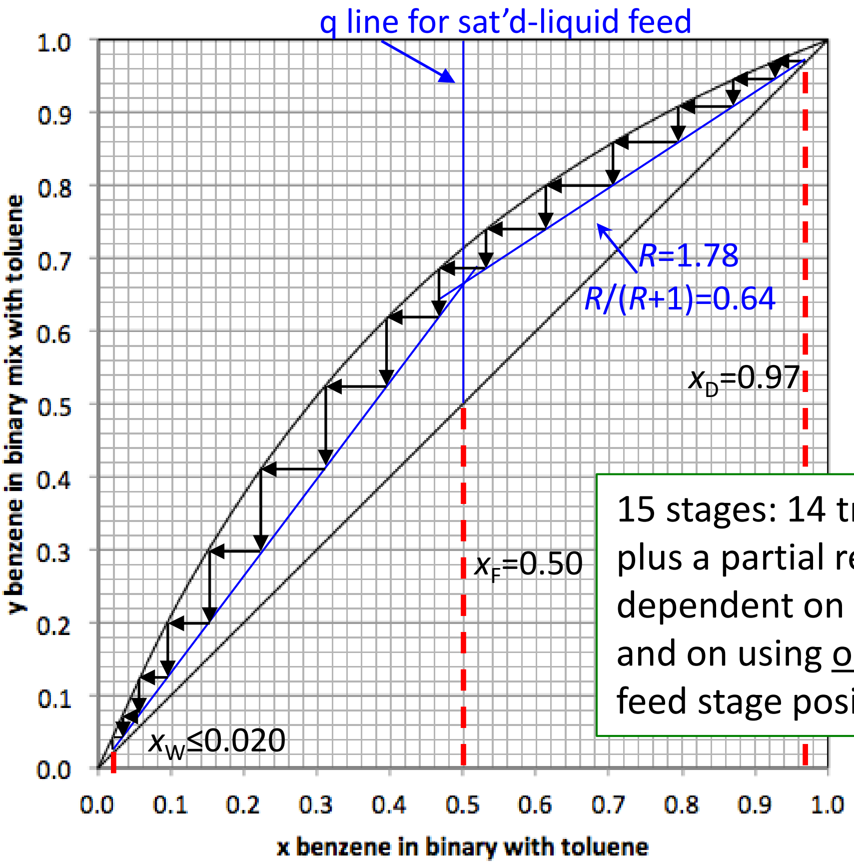
With *R*, solve for equilibrium, then the balance; then equilibrium, then balance, …

Let’s design a column: N, R, n_{feed}.

Here is an equilibrium graph, or use α_{AB}=2.485.

Need feed and product specs:
x_F=0.50, saturated liquid;
x_D=0.97 (variable?);
x_W≤0.020

Reflux ratio 1.78 (but *R&q* or *R&N* or *q&N* could be design variables)



15 stages: 14 trays plus a partial reboiler; dependent on *R*, *q*, and on using optimal feed stage position

Use graphical method OR stepwise calcs.

$x_D=x_0=$	
x_1 ← equil	y_1
x_2 balance	y_2
x_3	y_3
x_4	y_4
x_5	y_5
x_6	y_6
x_7	y_7
x_8	y_8
x_9	y_9
x_{10}	y_{10}
x_{11}	y_{11}
x_{12}	y_{12}
x_{13}	y_{13}
x_{14}	y_{14}

Three key factors to note:

- A vaporization with a condenser (one equilibrium stage) can do a lot of separation if relative volatility is high, but ...
- Countercurrent contacting is crucial for purification.
  - One stream gets higher in concentration, one gets lower; maximize the gradient.
- A lot of heat/cooling and heat transfer is required.
  - Working against entropy costs money.
  - Compute the required cooling for condensing the vapor flow V_1.
  - Compute heating for a boil-up rate V_N in the reboiler, plus heating feed to the desired q state.

Could you do it more precisely or different ways?

- Could you start from the bottom and solve upwards?
- Could you use a more exact equilibrium calculation?
- Could you start from the top and the bottom and work to the feed stage?
- Could you include a side-stream draw?
- Could you calculate changes in L and V (and L/V) as you go?
- Could you solve for multicomponent mixtures?
- Could you include reaction from having a catalyst on one stage?

1982 Eastman Chemical Co. patent by Victor H. Agreda (NCSU BS '75, MS '77, PhD '79)

Reactive distillation process for the production of methyl acetate

US4435595A
United States

Download PDF | Find Prior Art | Similar

Inventor: Victor H. Agreda, Lee R. Partin

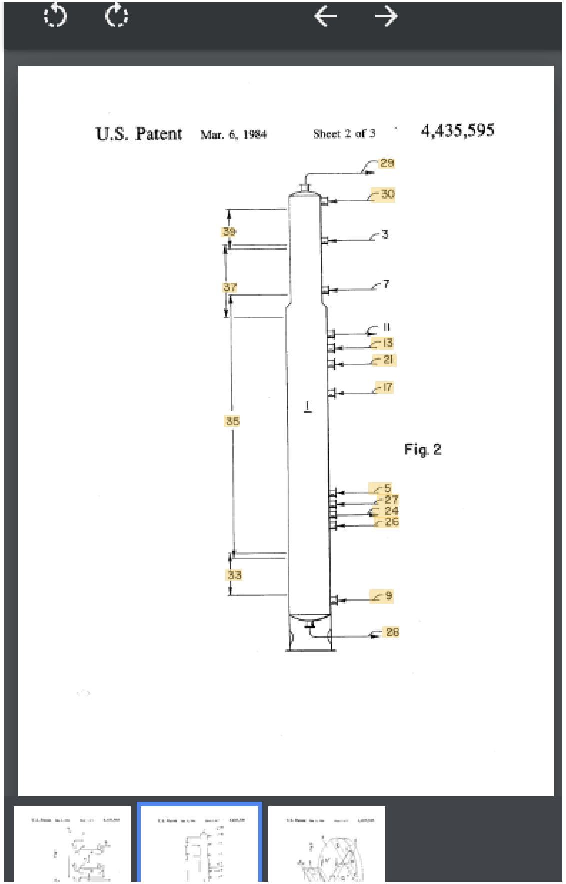
Current Assignee : Eastman Chemical Co

Worldwide applications

1982 • ~~US~~ 1983 • ~~JP~~ ~~DE~~ ~~EP~~ WO ~~CA~~ IT ZA

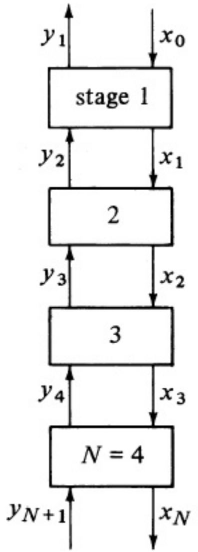
Application US06/371,626 events

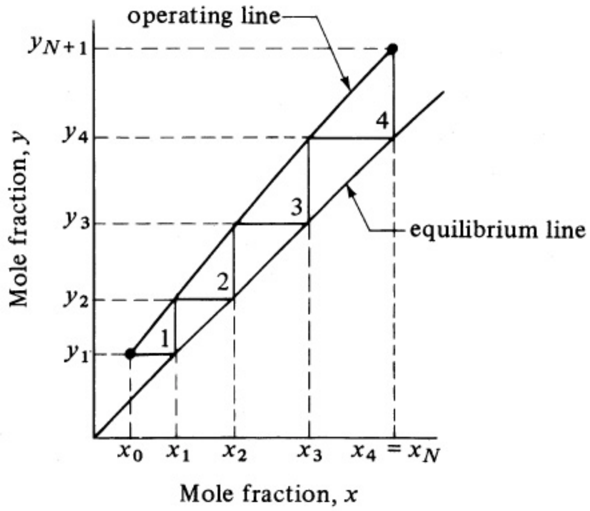
1982-04-26	Application filed by Eastman Kodak Co
1982-04-26	Priority to US06/371,626
1983-12-16	Assigned to EASTMAN KODAK COMPANY ROCKESTER NY A CORP
1984-03-06	Application granted
1984-03-06	Publication of US4435595A
1994-08-30	Assigned to EASTMAN CHEMICAL COMPANY
2002-04-26	Anticipated expiration
Status	Expired - Lifetime



You can design multi-stage absorption/stripping.

- A liquid stream and a gas stream with a third component.
- Suppose you have PFAS in a air stream and want to trap it in a liquid solvent.
    - Use $y_{i,n}$ and V_n for PFAS in the gas stream (feed).
    - Use $x_{i,n}$ and L_n for PFAS in the solvent stream.
- As before, apply the stage *n* subscript to the streams <u>leaving</u> each stage, presumably in equilibrium (maybe Henry's Law $y_{i,n} = m\, x_{i,n}$).
- Then, just as before: Successively apply equilibrium relation, material balance, equilibrium relation, material balance...
- What is different? Not thermally driven!





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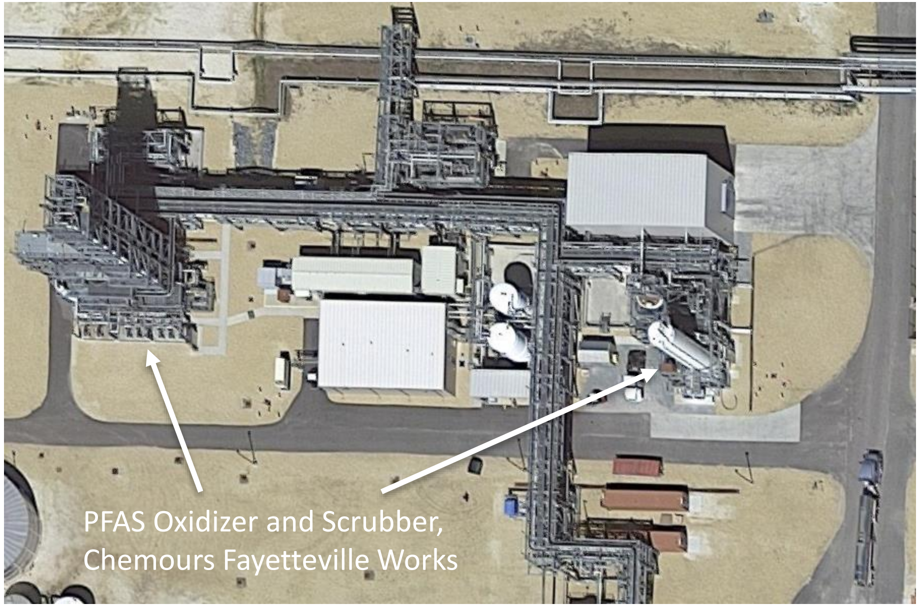
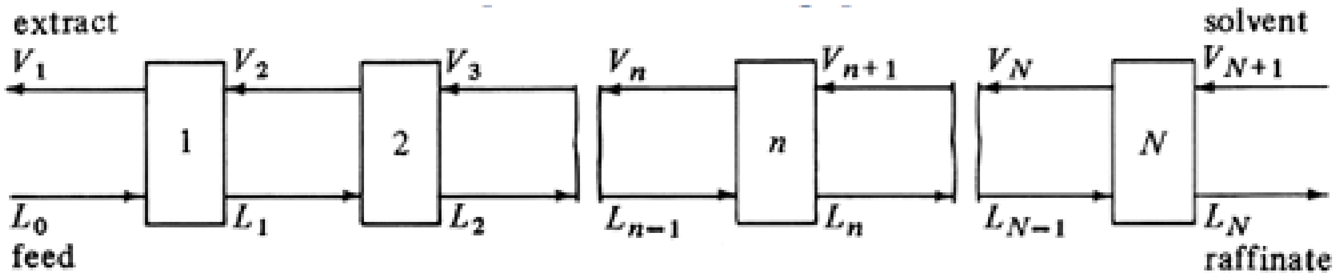


Usually not staged but continuous counter-contacting, rate-limited by mass transfer.

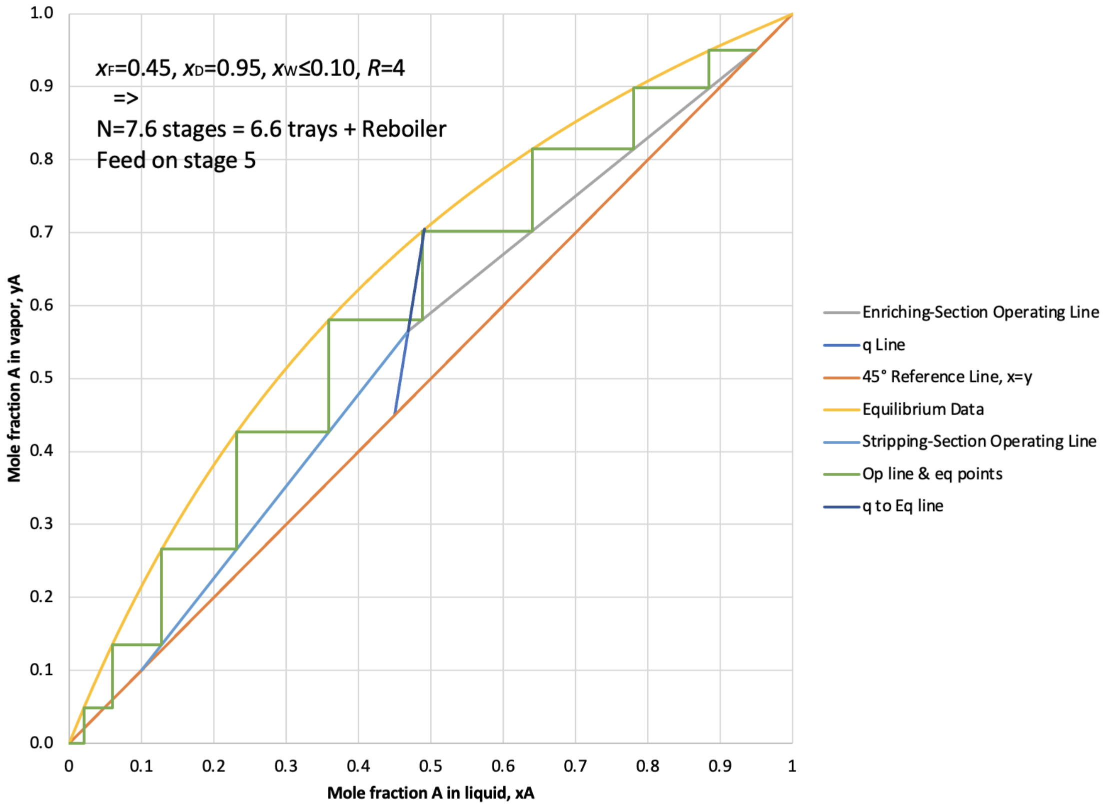
Could you design multi-stage extraction of an amino acid?



- Two **liquid** streams with a amino acid transferring from one phase to another:
  - Use "$y_{i,n}$" and "V_n" for the solvent-rich "extract" stream (here, right to left).
  - Use $x_{i,n}$ and L_n for the feed-to-"raffinate" stream (here, left to right).
- As before, apply the stage *n* subscript to the streams <u>leaving</u> the stage, presumably in equilibrium ($\boldsymbol{\gamma_{yA} y_{A,n} = \gamma_{xA} x_{A,n}}$).
- Then, just as before: Successively apply equilibrium relation, material balance, equilibrium relation, material balance...
- What is different?

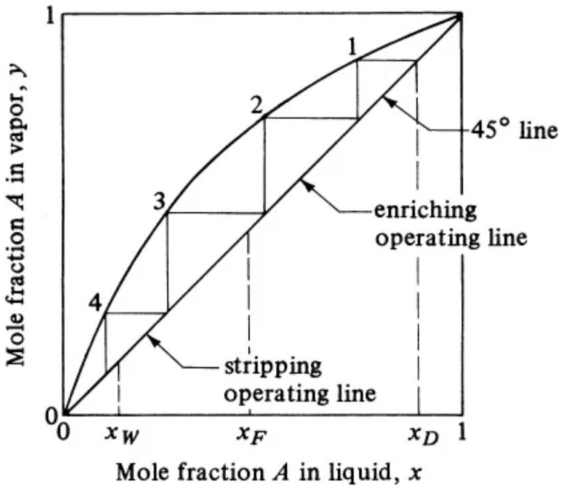
We can examine design sensitivities.

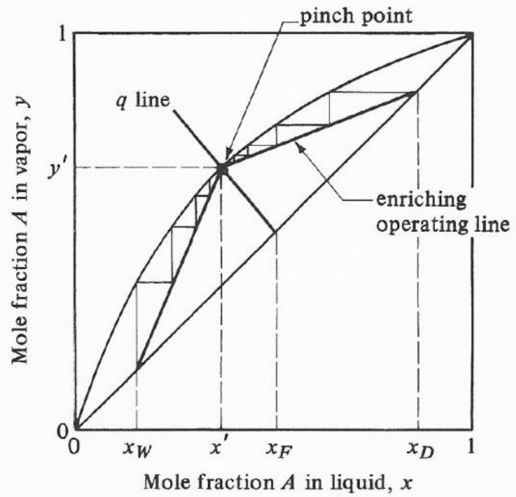
- Tighten or loosen distillate and/or bottoms specification.
- Change quality of feed (q) or its composition.



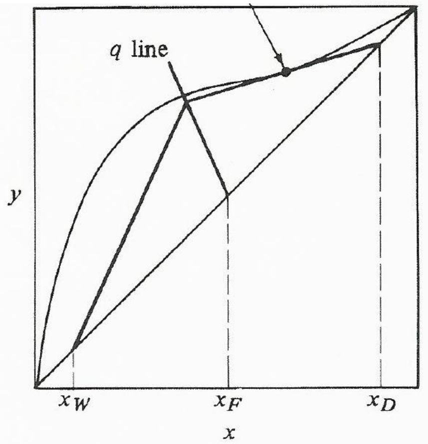
We can examine design sensitivities.

- Dial reflux ratio up and down:
  - "Total reflux" or $R \rightarrow \infty$ gives a lower bound for number of stages (minimum number of stages).
  - "Pinch point" determines R_{min} (infinite number of stages required), determines R (optimal number of stages, ~1.5 R_{min})….>





"tangent pinch"



We have L/V but need actual flowrates to determine utilities and column diameter.

How to get reflux rate, L_0?

How to get overhead vapor rate, V_1?

How to get boil-up rate, V_{N+1}?

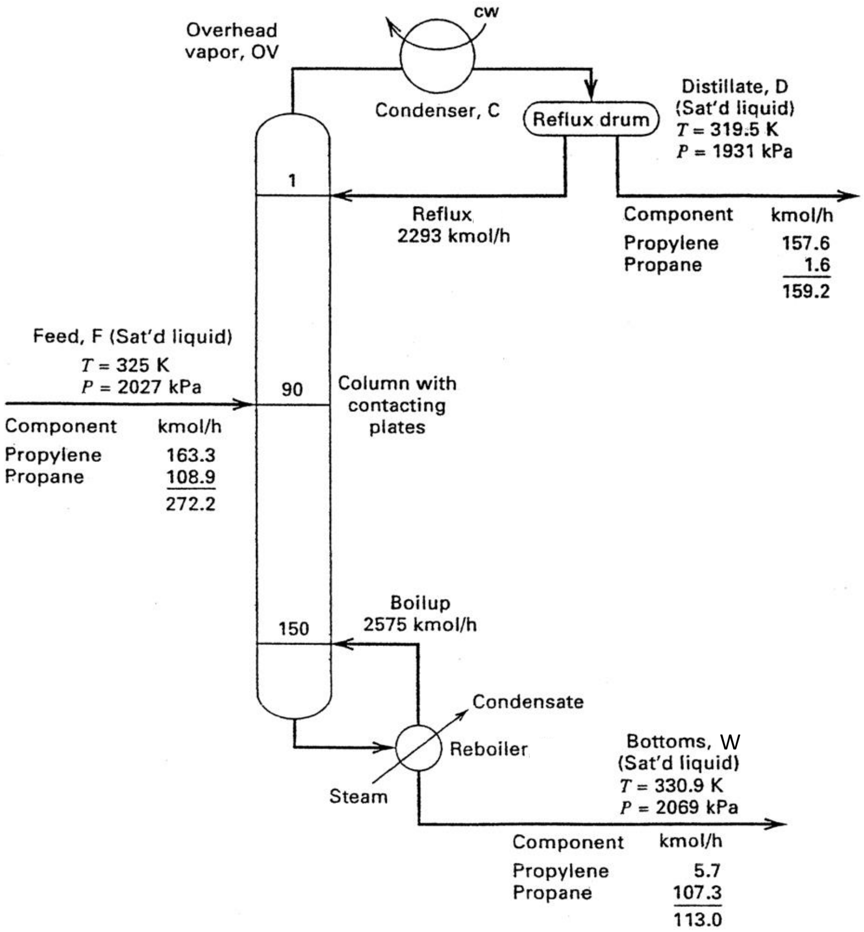


Fig. 1.16; Seader, Henley, and Roper, *Separation Process Principles*, 3rd Ed.

How to get reflux rate, L_0?
Reflux Ratio*D

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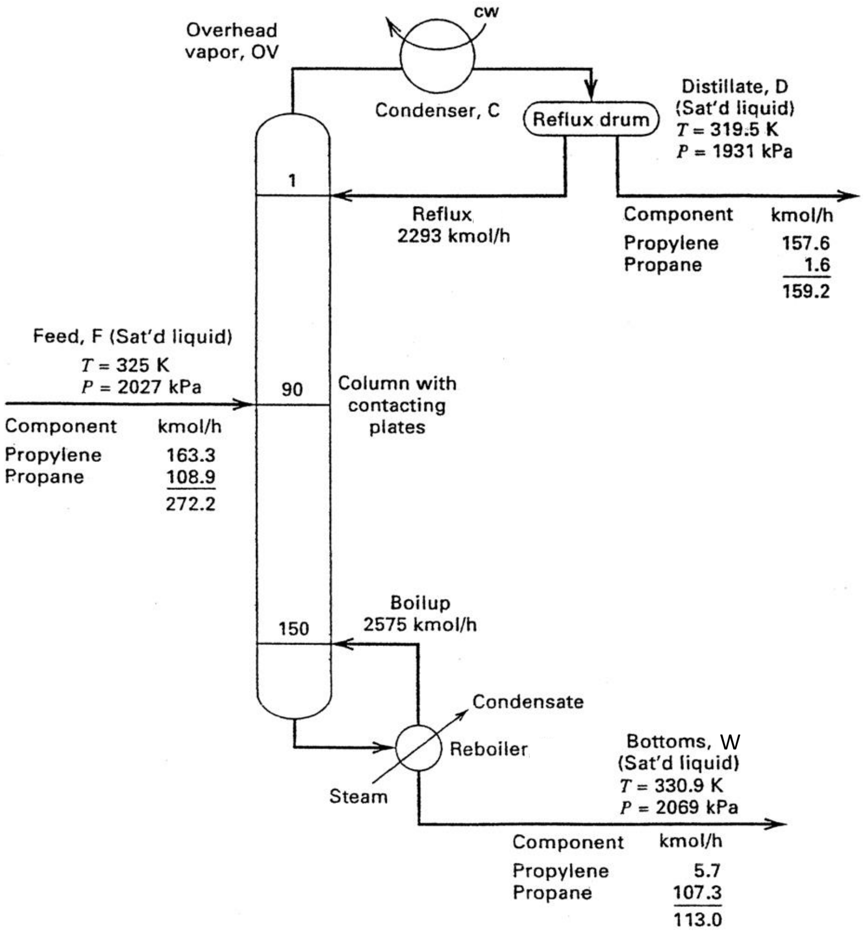

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How to get reflux rate, L_0?
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How to get overhead vapor rate, V_1?
L_0+D

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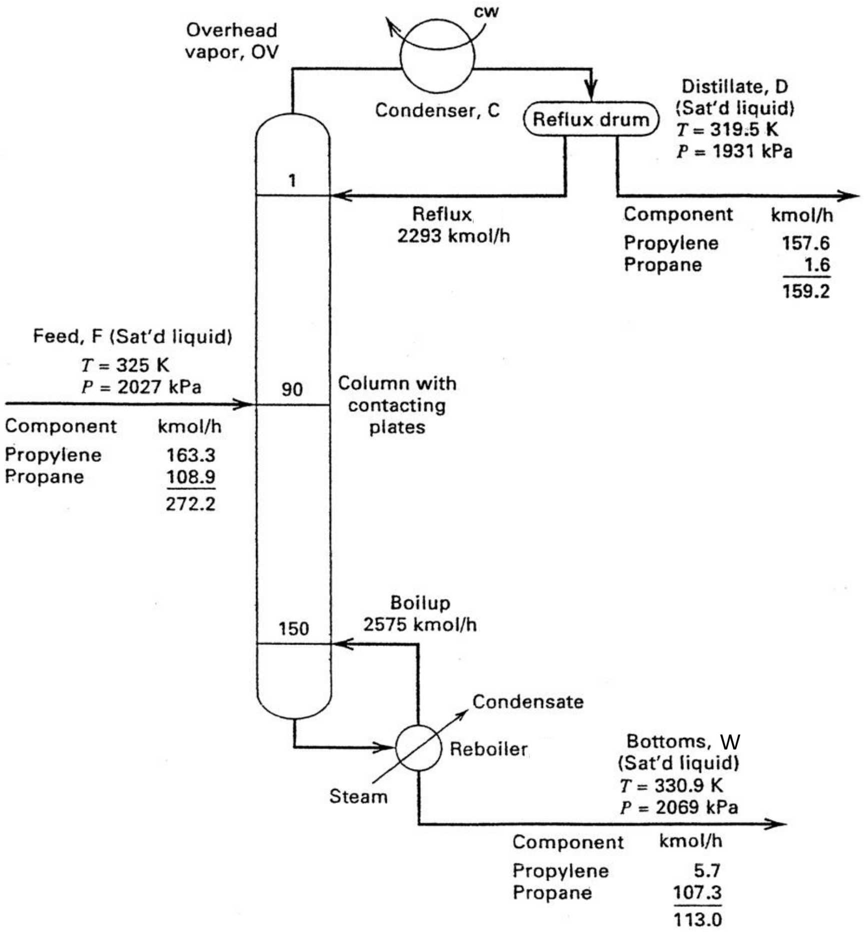

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How to get reflux rate, L_0?
Reflux Ratio*D

How to get overhead vapor rate, V_1?
L_0+D

How to get boil-up rate, V_{N+1}?
If R and q are set, V_{N+1} must conform to the Stripping Section Operating Line (the balance).

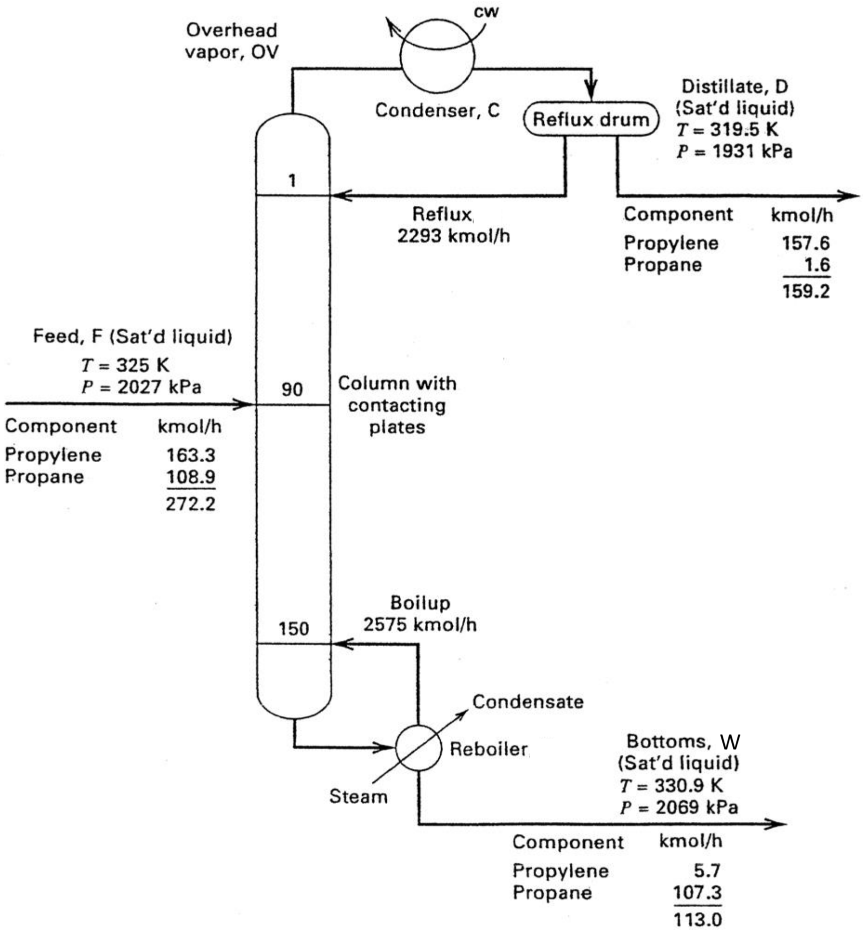
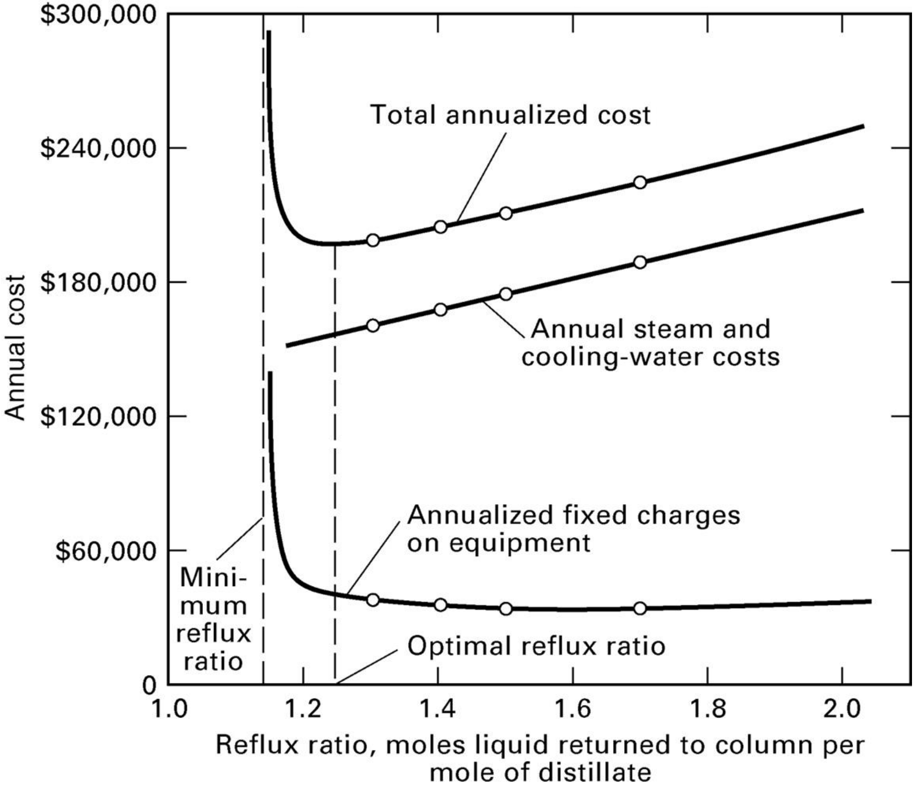


Fig. 1.16; Seader, Henley, and Roper, *Separation Process Principles*, 3rd Ed.

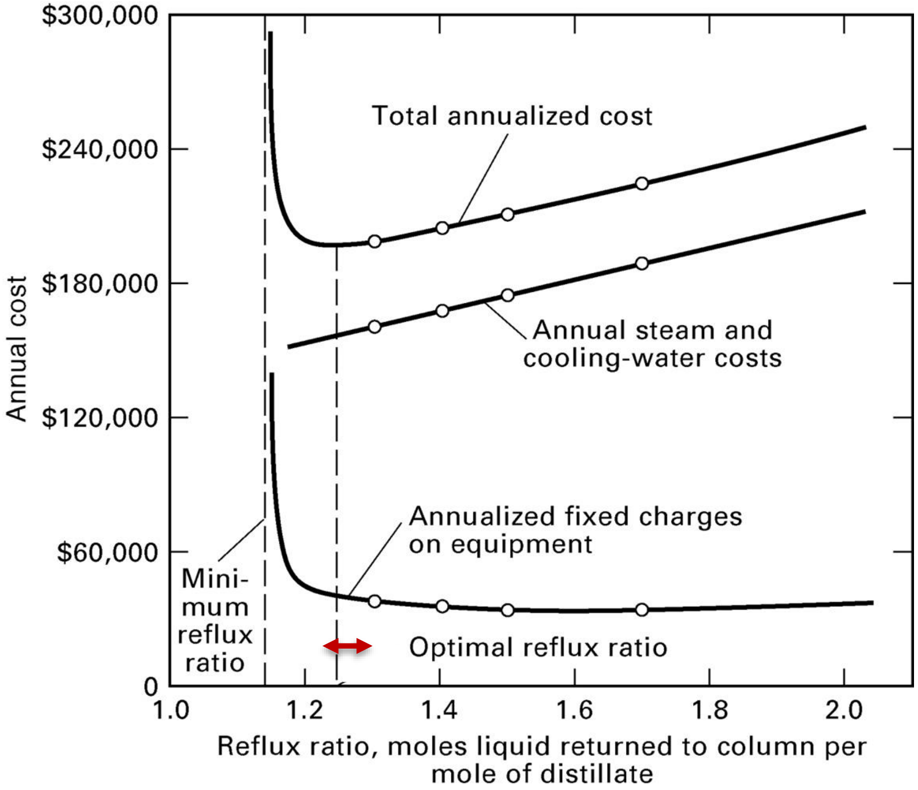
Expensive capital (CAPEX), but utilities are expensive, too.

Here is one case:



How to calculate utilities costs?

Utilities are the most expensive operating cost (OPEX).



Heat required for boil-up is

$$V_N \cdot \lambda_W = \dot{m}_{steam}\lambda_{steam},$$

and heat for heating the feed is

$$qF \cdot \lambda_F = \dot{m}_{st}\lambda_{steam}$$

Cooling req'd for the condenser is

$$V_1 \cdot \lambda_1 = D(R+1)\,\lambda_1 = \dot{m}_{cw}C_{p,cw}(T_o - T_{in})$$

Note:

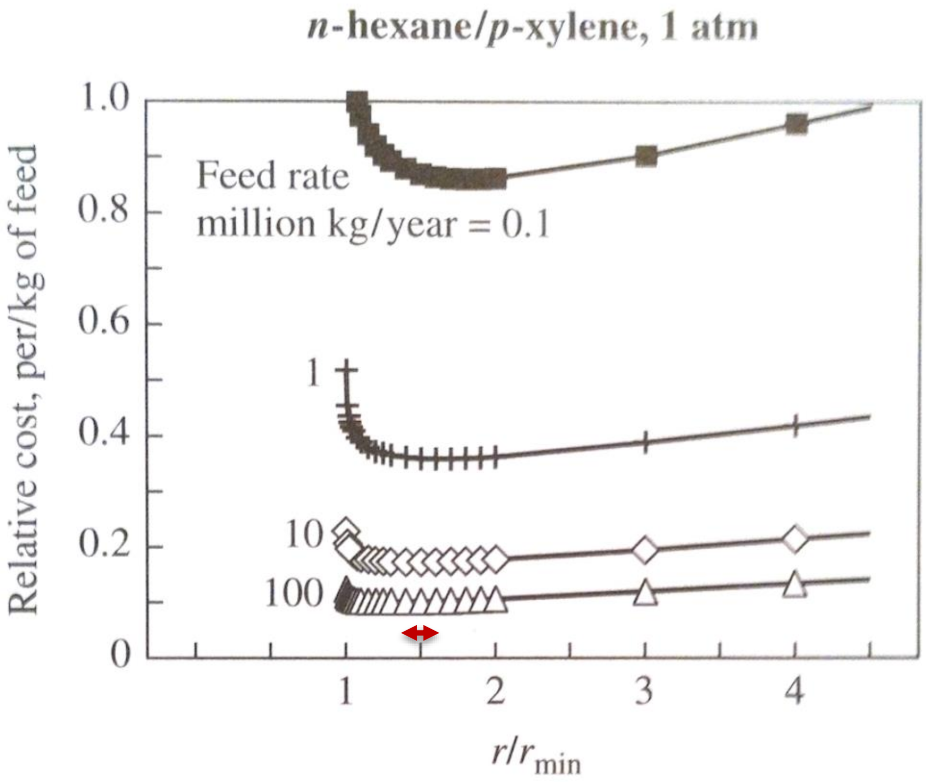
1. If feed is sat'd liquid,
$$Q_{reboiler} \cong Q_{condenser}$$
2. For ideal solution,
$$\lambda_{mix} = x_A\lambda_A + x_B\lambda_B$$

Here, optimal $R \cong 1.25$, divided by $R_{min} \cong 1.13$, is 1.1

Consider the numbers for C_3H_6/C_3H_8.

1. Find the flow rate of vapor going through the distillate (V_1) and the flow rate of boil-up (V_{N+1}).
   a. V_1=2293+159 kmol/hr of ~propene (99%); $\Delta H_{vapn} \cong$ 18.9 kJ/mol at 320 K (NIST Webbook)
   b. V_{N+1}=2575 kmol/hr of ~propane (95%); $\Delta H_{vapn} \cong$ 18.7 kJ/mol at 330 K
2. Determine the enthalpy removed from the distillate and added to the reboiler, $\dot{m}\lambda$.
   a. 46•10^6 kJ/hr CW and 48•10^6 kJ/hr steam
3. Find the cooling water and steam required to complete the energy balance.
   a. $Q_{cw} = \dot{m}_c C_p \Delta T$ so $\dot{m}_c = \frac{Q_{cw}}{C_p \Delta T} = \frac{46E6\ kJ/h}{\left(4.2\frac{kJ}{kg\bullet K}\right)(50-20°C)} = 3.7E5\ kg/h$
   b. $Q_{st} = \dot{m}_{st}\lambda_{H2O}$ so $\dot{m}_{st} = \frac{Q_{st}}{\lambda_{H2O}} = \frac{48E6\ kJ/h}{2257\frac{kJ}{kg}} = 21E3\ kg\ steam/h$
4. Plug into a cost equation and find the cost required to run.
   a. Estimate \$3/1000 gal CW=\$0.0008/kg so \$290/h or \$2.6 million/yr CW
   b. Estimate \$14/GJ steam “loaded cost” (fuel + generation) so \$670/h or \$6 million/yr steam

Detailed designs show that R (optimal number of stages) $\cong 1.5R_{min}$



Fenske equation for stages: Useful and dangerous.

- Requires constant relative volatility; may be very wrong if nonideal mixtures.
- Requires total condenser and assumes optional feed location.

$$N_{min,Fenske} = \frac{\log\left(\frac{X_D}{1-X_D}\frac{1-X_w}{X_w}\right)}{\log \alpha_{ave}}$$

where $\alpha_{ave} = \sqrt{\alpha_{top}\alpha_{bottoms}}$

- In this class, assume you have to determine stages from balance equations unless I tell you to use Fenske – or if you're checking yourself.